Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Substitution: $a = \frac{(25-0)}{20}$ (1) = 1.25 [m/s^2] (1) Substitution into $F = ma$ i.e. 5 000 [N] = $m \times 1.25$ [m/s^2] (**ecf**) (1) = 4 000 [kg] (1) **Alternative** Substitution for v and t values from the graph: $F = \frac{\Delta p}{t}$ so $F = \frac{m(25-0)}{20}$ (2) [inclusion of m (1), substitution of values (1)] 5000 [N] = $m \times 1.25$ [m/s^2] **ecf** (1) m = 4 000 [kg] (1) NB. Ecf on acceleration only if it is calculated [even mysteriously losing a '–' sign] Penalise mistakes in sign by – 1.	1 1	1 1		4	4	
	(b)			Acceleration would be less [as mass increased] (1) [Not: slower] So the gradient would be less steep [Not: the line AB would be longer] (1)	1	1		2	1	
	(c)	(i)		If the [external] forces on an object are balanced/zero <u>resultant</u> force [accept: equal and opposite] (1) the object moves at constant velocity [or speed in a straight line] or is stationary / acceleration zero / uniform motion in a straight line (1)	2			2		
		(ii)		The velocity [accept: speed] is constant / is 25 m/s [accept: it has reached its terminal velocity / it is travelling at terminal velocity] (1) <u>Resultant</u> force [on bus] = zero / forces [on bus] are balanced (1)	1	1		2	1	
	(d)			Times at 12.5 m/s : 10 ± 2 s and 170 ± 2 s (1) [or by implication] 170 – 10 = 160 [s] (1) [accept 156 to 164 s]	1	1		2	1	

Question	Marking details	AO1	AO2	AO3	Total	Maths	Prac
		Marks available					
(e)	Total area under graph = total distance travelled by bus (1) [Award on attempt to calculate total area under graph, not 25 × 190 ! : use of trapezium rule, or multiple areas] Substitution: (0.5 × 20 × 25) + (25 × 130) + (0.5 × 40 × 25) Or 0.5 × (130 + 190) × 25 (1) = 4 000 [m] (1) Other distances between 3900 m and 4 100 m → 2 marks up to this point] $v = \frac{4\,000(\textbf{ecf})}{190}$ = 21.1 [m/s] (1) [20.5 – 21.6 m/s from above] Horizontal line at 21 m/s **ecf** (1) Tolerance ± ½ square. Ignore any parts of the line with t > 190 s & don't penalise slightly shorter lines **Alternative:** Use of equations of motion: $x = \frac{(v+u)t}{2} + (vt) + \frac{(v+u)t}{2}$ (1) Substitution: $x = \frac{(25+0)}{2}$ x 20 +(25 ×130) + $\frac{(0+25)}{2}$ x 40 (1) = 4 000 [m] (1) Other distances between 3900 m and 4 100 m → 2 marks up to this point] $v = \frac{4\,000(\textbf{ecf})}{190}$ = 21.1 [m/s] (1) Horizontal line at 21 m/s **ecf** (1) Tolerance ± ½ square. Ignore any parts of the line with t > 190 s & don't penalise slightly shorter lines	1 1	 1 1 1		5	4	
	Question 4 total	**9**	**8**	**0**	**17**	**11**	**0**